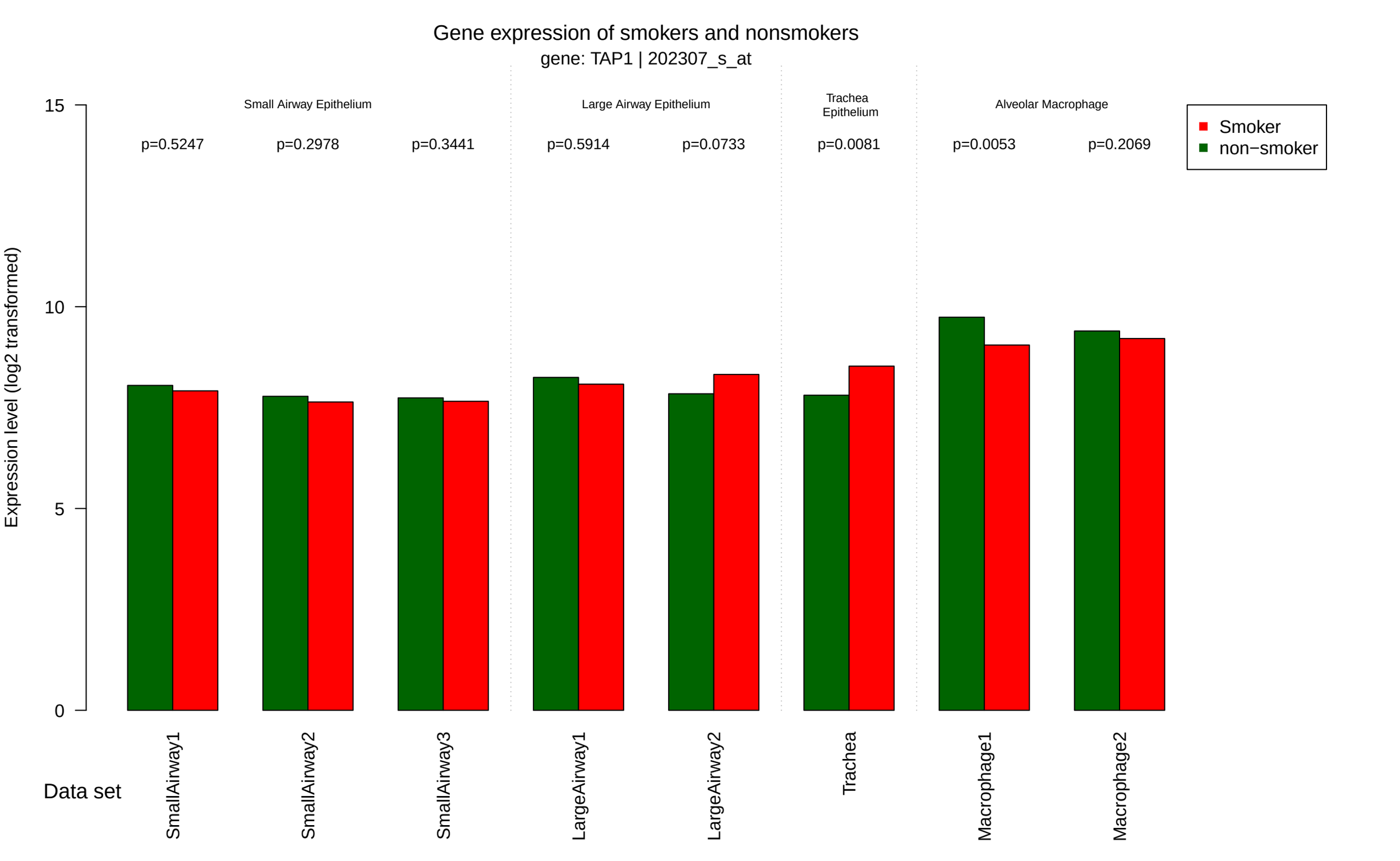
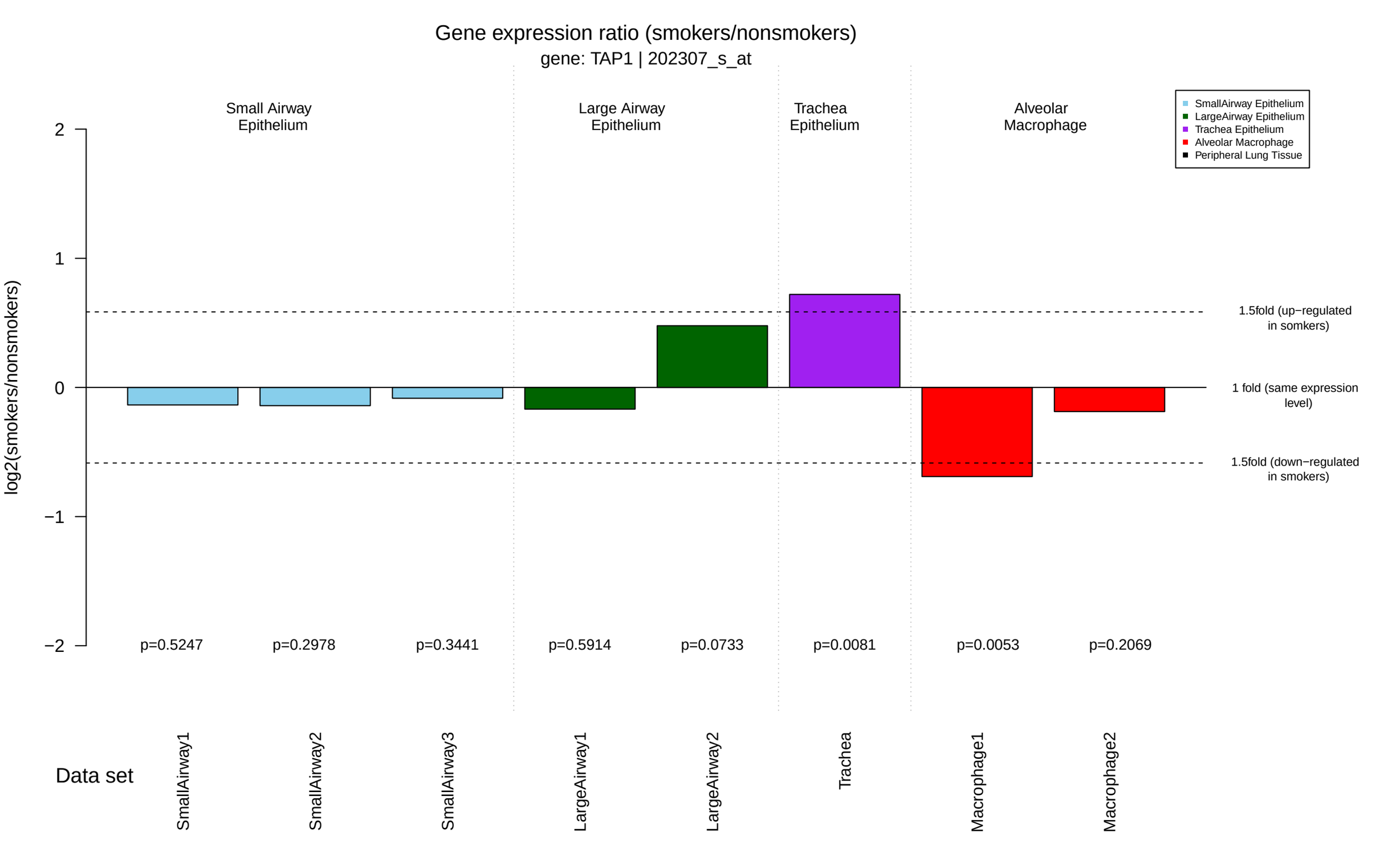
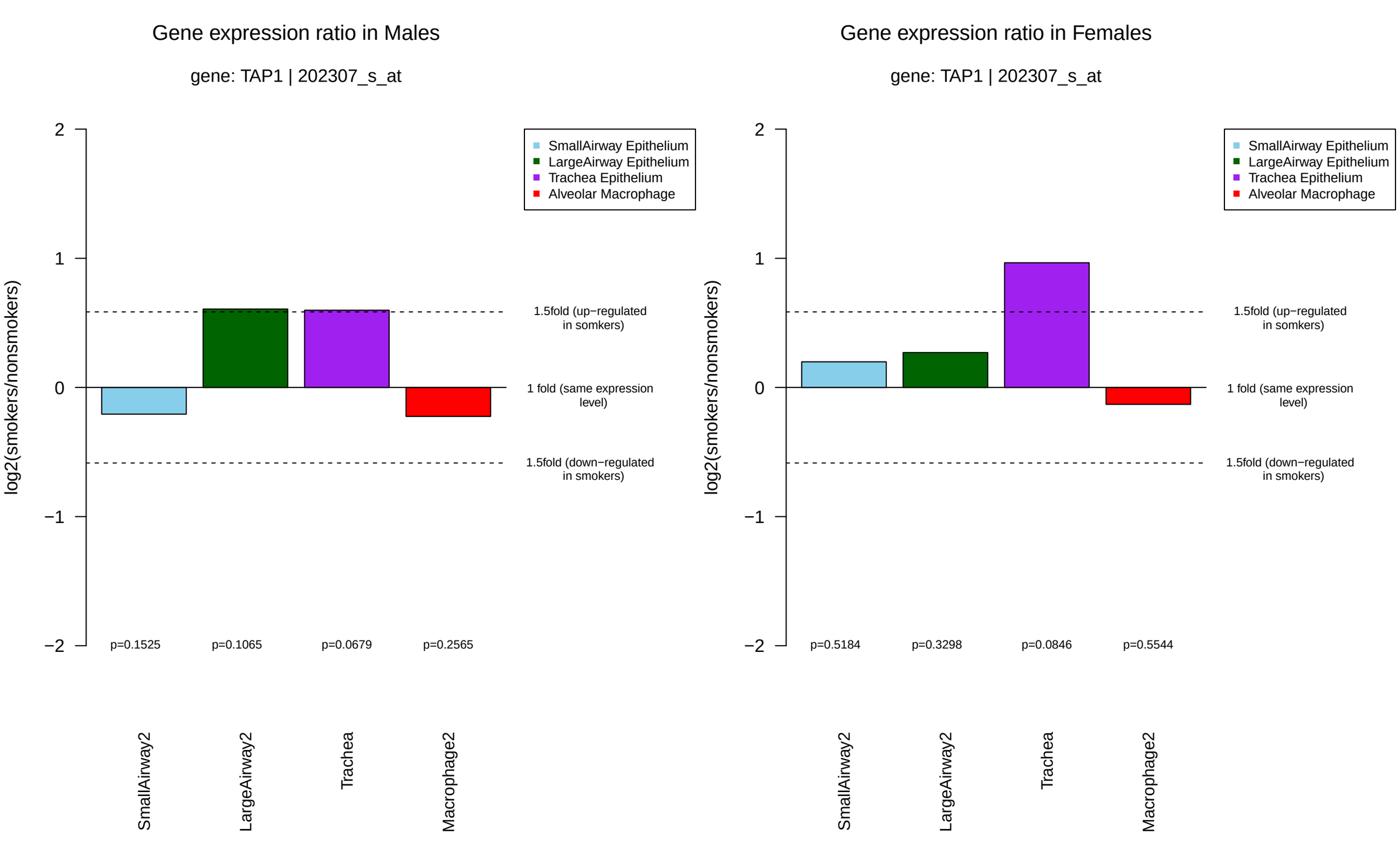
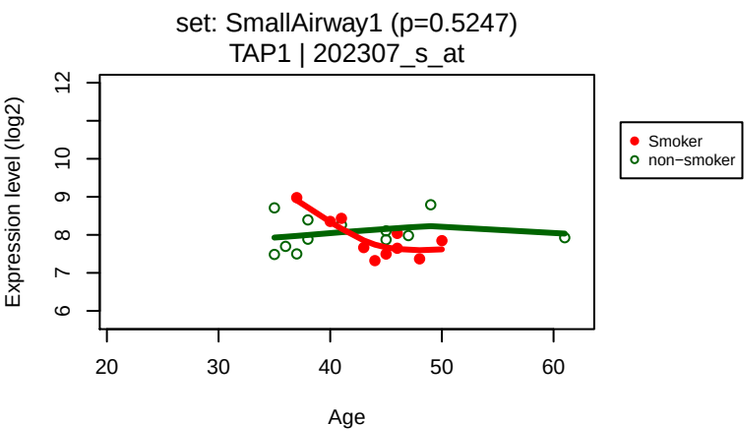
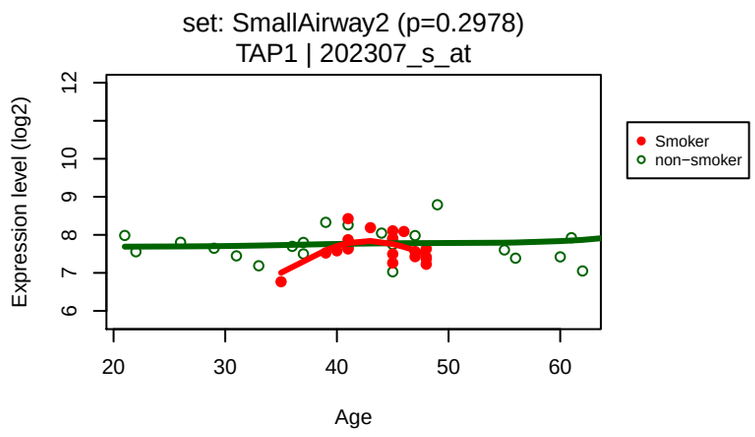
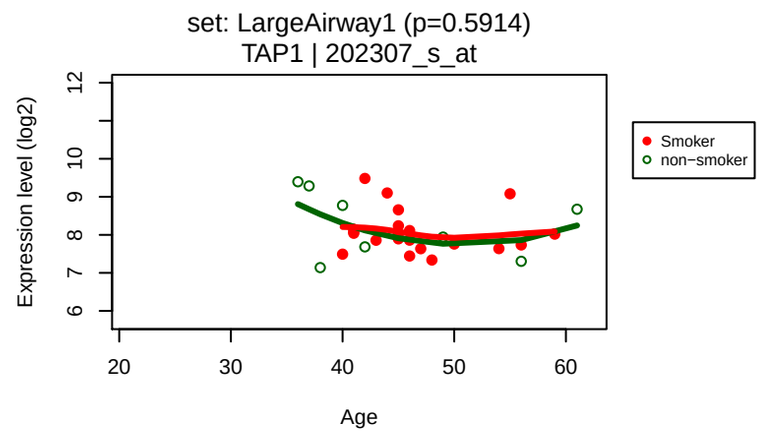

Gene expression of smokers and nonsmokers
gene: TAP1 | 202307_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5247
p=0.2978
p=0.3441
p=0.5914
p=0.0733
p=0.0081
p=0.0053
p=0.2069
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

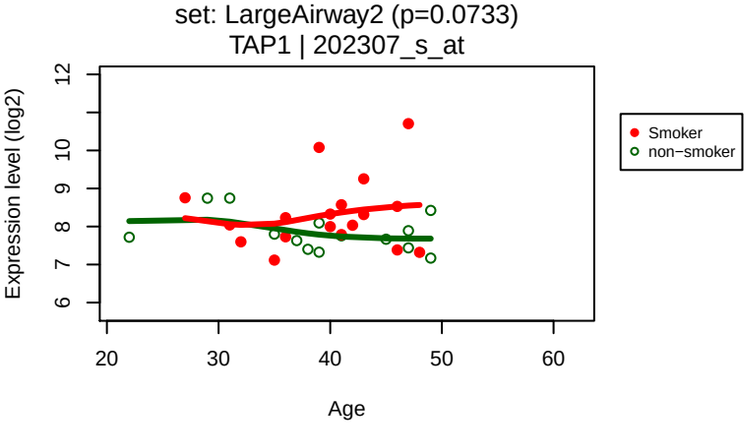

Gene expression ratio (smokers/nonsmokers)
gene: TAP1 | 202307_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5247
p=0.2978
p=0.3441
p=0.5914
p=0.0733
p=0.0081
p=0.0053
p=0.2069
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

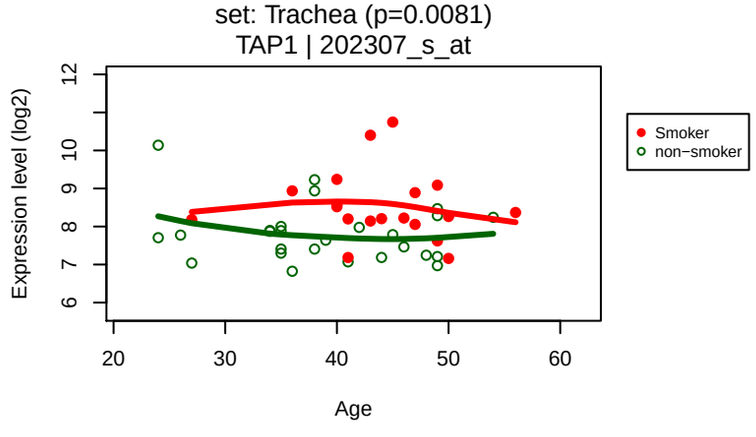

Gene expression ratio in Males
gene: TAP1 | 202307_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1525
p=0.1065
p=0.0679
p=0.2565
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TAP1 | 202307_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5184
p=0.3298
p=0.0846
p=0.5544
SmallAirway2
LargeAirway2
Trachea
Macrophage2

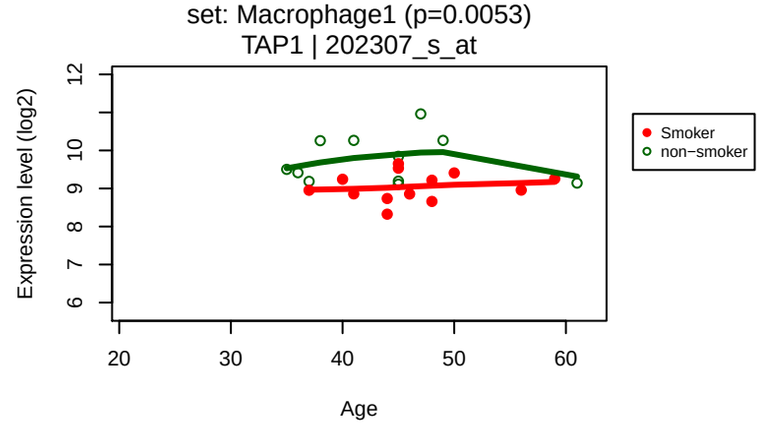

set: SmallAirway1 (p=0.5247)
TAP1 | 202307_s_at
Expression level (log2)
Age
Smoker
non−smoker

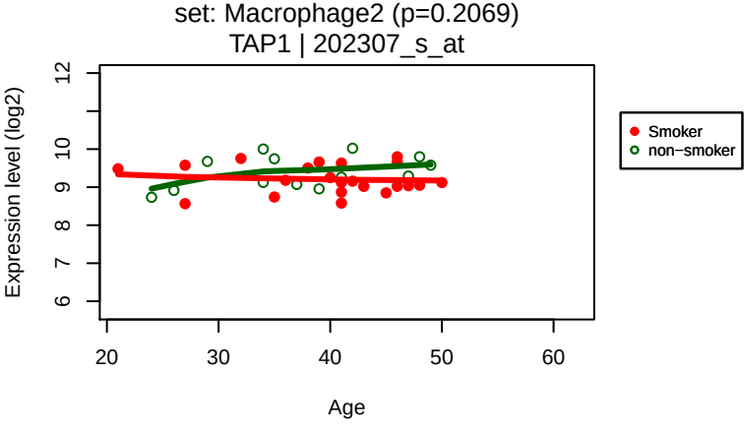

set: SmallAirway2 (p=0.2978)
TAP1 | 202307_s_at
Expression level (log2)
Age
Smoker
non−smoker

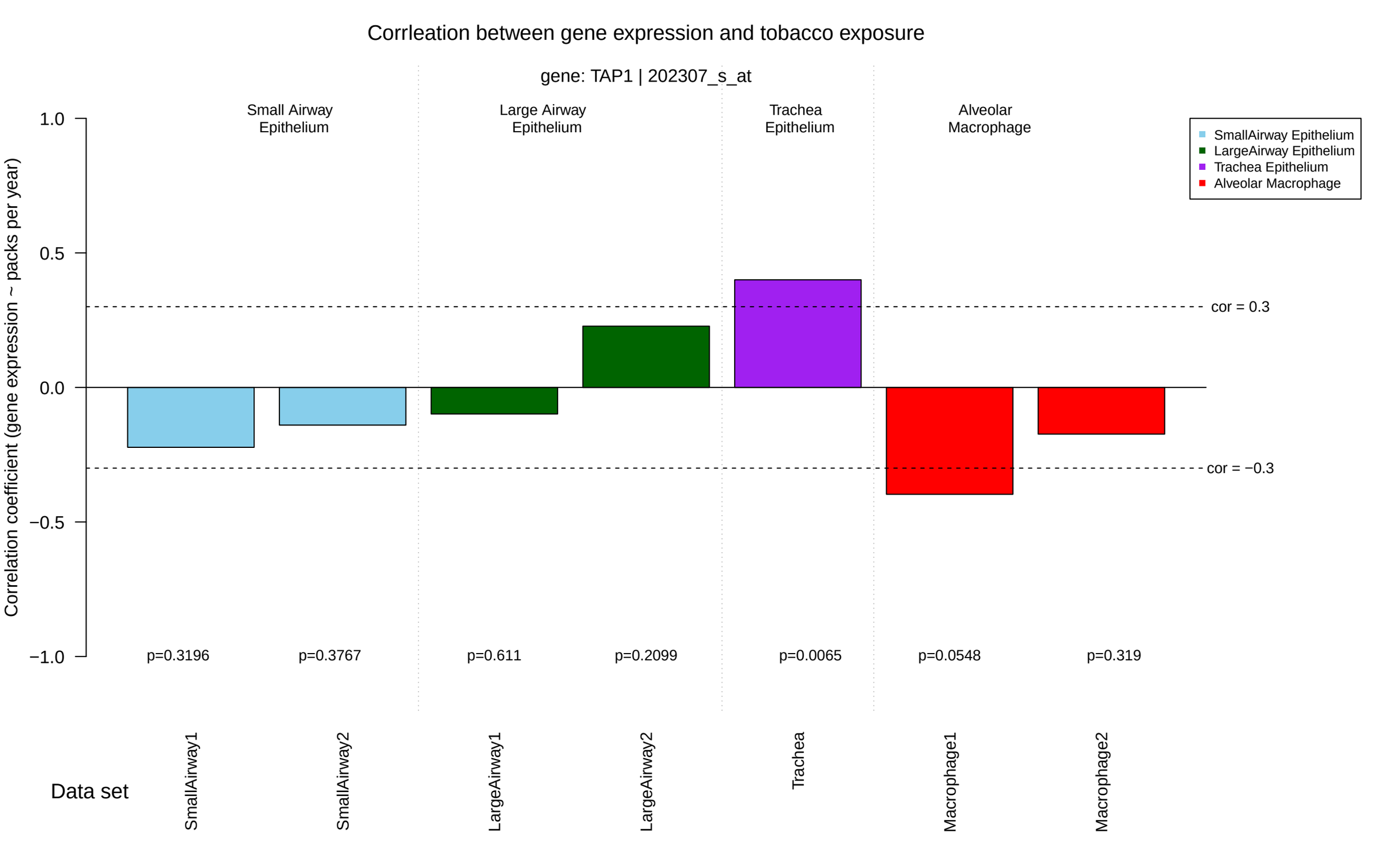

set: LargeAirway1 (p=0.5914)
TAP1 | 202307_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0733)
TAP1 | 202307_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0081)
TAP1 | 202307_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0053)
TAP1 | 202307_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2069)
TAP1 | 202307_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TAP1 | 202307_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3196
p=0.3767
p=0.611
p=0.2099
p=0.0065
p=0.0548
p=0.319
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2